

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE CODE: Data Warehousing and Data Mining

COURSE NAME: CSA1603

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

Assessment Tool Weightage: 50

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

Criteria	Understanding of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total(10)
Weightage	25%	30%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I P. GAYATHRI/192425277_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

P. GAYATHRI

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

1.Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Photograph:

Age of child	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

ANS:

Covariance and Correlation

The question gives the following preferences for photographs A, B and C and asks for covariance and correlation using cov() and cor().

Program R Code

```
# Q1 - Covariance and Correlation
```

```
A <- c(18, 2, 20)
```

```
B <- c(22, 28, 10)
```

```
C <- c(20, 40, 40)
```

```
data1 <- data.frame(A, B, C)
```

```
# Display data
```

```
data1
```

```
# Covariance between B and C
```

```
cov(B, C)
```

```
# Covariance matrix
```

```
cov(data1)
```

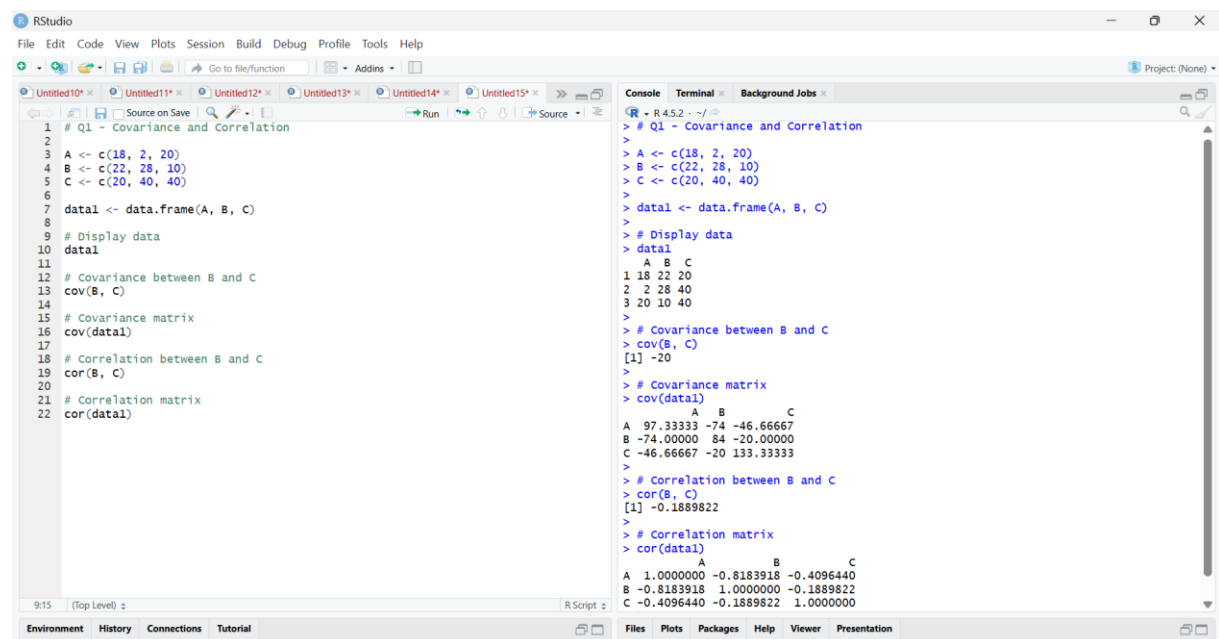
```
# Correlation between B and C
```

```
cor(B, C)
```

```
# Correlation matrix
```

```
cor(data1)
```

Output:



```
# Q1 - Covariance and Correlation
1
2
3 A <- c(18, 2, 20)
4 B <- c(22, 28, 10)
5 C <- c(20, 40, 40)
6
7 data1 <- data.frame(A, B, C)
8
9 # Display data
10 data1
11
12 # Covariance between B and C
13 cov(B, C)
14
15 # Covariance matrix
16 cov(data1)
17
18 # Correlation between B and C
19 cor(B, C)
20
21 # Correlation matrix
22 cor(data1)
```

```
> # Q1 - Covariance and Correlation
>
> A <- c(18, 2, 20)
> B <- c(22, 28, 10)
> C <- c(20, 40, 40)
>
> data1 <- data.frame(A, B, C)
>
> # Display data
> data1
  A B C
1 18 22 20
2  2 28 40
3 20 10 40
>
> # Covariance between B and C
> cov(B, C)
[1] -20
>
> # Covariance matrix
> cov(data1)
      A      B      C
A 97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
>
> # Correlation between B and C
> cor(B, C)
[1] -0.1889822
>
> # Correlation matrix
> cor(data1)
      A      B      C
A 1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
```

Result

The sample covariance between B and C is **-20**, and the sample correlation is approximately **-0.189**. The covariance and correlation matrices were successfully calculated.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID Player Name Points Scored

P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10

Player ID Player Name Points Scored

P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

ANS:

Boxplot and Outlier Detection

The question gives points scored by 15 tennis players and asks how a coach can use a boxplot to identify outliers.

R Code:

```
# Q2 - Boxplot and Outlier Detection
```

```
points <- c(12, 15, 14, 10, 18, 20, 22, 13,  
            11, 35, 16, 17, 19, 21, 23)
```

```
# Summary
```

```
summary(points)
```

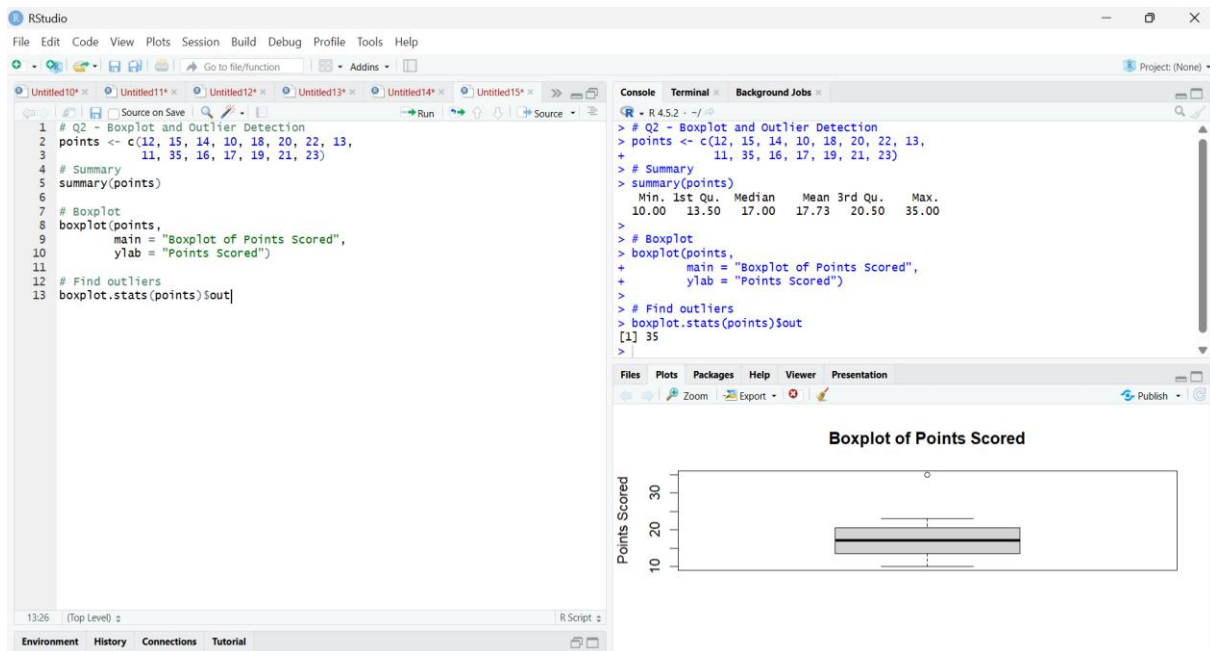
```
# Boxplot
```

```
boxplot(points,  
         main = "Boxplot of Points Scored",  
         ylab = "Points Scored")
```

```
# Find outliers
```

```
boxplot.stats(points)$out
```

Output:



Result:

Player J, who scored **35 points**, is an outlier.

The boxplot shows that most players scored between approximately **10 and 23 points**, while 35 is unusually high.

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set

young	high	employed	good	yes
young	high	self-employed	average	yes
middle	high	employed	good	yes
old	medium	employed	good	yes
old	low	unemployed	poor	no
old	low	self-employed	average	no
middle	low	unemployed	poor	no
young	medium	employed	average	yes
young	low	unemployed	poor	no
old	medium	self-employed	good	yes
middle	medium	employed	average	yes

middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

ANS:

Decision Tree Classifier

The question asks you to use historical customer information to predict **Loan Approval (Yes/No)** using a Decision Tree. The dataset contains 15 records.

R Code:

```
data <- data.frame(  
  Age = c("young","young","middle","old","old",  
          "old","middle","young","young","old",  
          "middle","middle","old","young","middle"),  
  
  Income = c("high","high","high","medium","low",  
             "low","low","medium","low","medium",  
             "medium","high","medium","medium","low"),  
  
  Employment = c("employed","self-employed","employed",  
                 "employed","unemployed","self-employed",  
                 "unemployed","employed","unemployed",  
                 "self-employed","employed","self-employed",  
                 "unemployed","self-employed","employed"),  
  
  Credit = c("good","average","good","good","poor",  
             "average","poor","average","poor","good",  
             "average","good","poor","average","poor"),  
  
  Loan = c("yes","yes","yes","yes","no",  
           "no","no","yes","no","yes",  
           "yes","yes","no","yes","no")  
)  
  
data[] <- lapply(data, factor)  
  
model <- rpart(  
  Loan ~ Age + Income + Employment + Credit,  
  data = data,  
  method = "class"  
)  
  
print(model)  
  
rpart.plot(model,  
  main = "Decision Tree for Loan Approval")
```

Output:

```

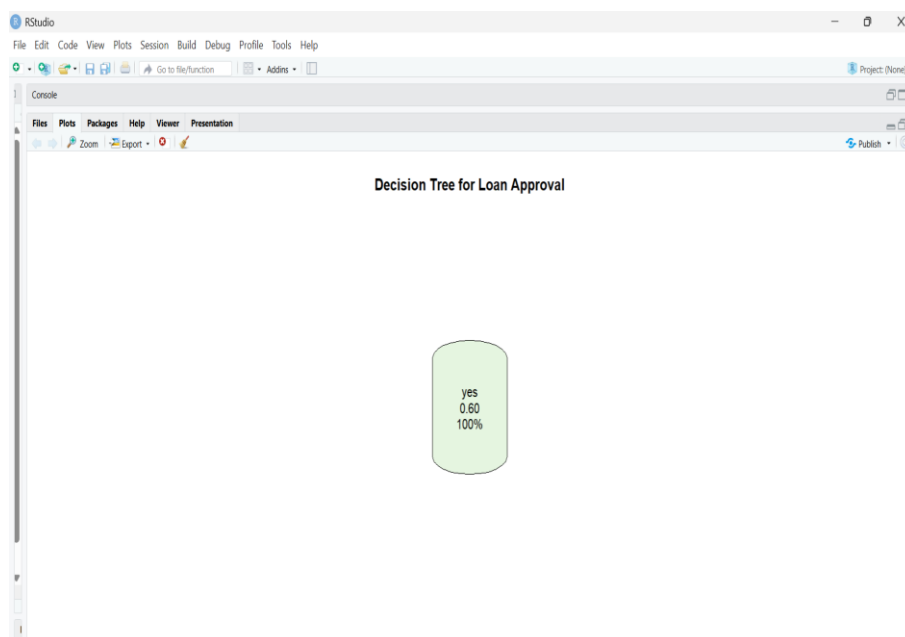
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source on Save Run Source
1 data <- data.frame(
2   Age = c("young", "young", "middle", "old", "old",
3         "old", "middle", "young", "young", "old",
4         "middle", "middle", "old", "young", "middle"),
5
6   Income = c("high", "high", "high", "medium", "low",
7             "low", "low", "medium", "low", "medium",
8             "medium", "high", "medium", "medium", "low"),
9
10  Employment = c("employed", "self-employed", "employed",
11                "employed", "unemployed", "self-employed",
12                "unemployed", "employed", "unemployed",
13                "self-employed", "employed", "self-employed",
14                "unemployed", "self-employed", "employed"),
15
16  Credit = c("good", "average", "good", "good", "poor",
17            "average", "poor", "average", "poor", "good",
18            "average", "good", "poor", "average", "poor"),
19
20  Loan = c("yes", "yes", "yes", "yes", "no",
21          "no", "no", "yes", "no", "yes",
22          "yes", "yes", "no", "yes", "no")
23 )
24
25 data[] <- lapply(data, factor)
26
27 model <- rpart(
28   Loan ~ Age + Income + Employment + Credit,
29   data = data,
30   method = "class"
31 )
32
33 print(model)
34
35 rpart.plot(model,
36   (Top Level) z
37 )
38
39
40
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43
44
45
46
47
48
49
50
51
52
53
54
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82
83
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85
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99

```

```

+ Employment = c("employed", "self-employed", "employed",
+               "employed", "unemployed", "self-employed",
+               "unemployed", "employed", "unemployed",
+               "self-employed", "employed", "self-employed",
+               "unemployed", "self-employed", "employed"),
+
+ Credit = c("good", "average", "good", "good", "poor",
+           "average", "poor", "average", "poor", "good",
+           "average", "good", "poor", "average", "poor"),
+
+ Loan = c("yes", "yes", "yes", "yes", "no",
+         "no", "no", "yes", "no", "yes",
+         "yes", "yes", "no", "yes", "no")
+ )
+ data[] <- lapply(data, factor)
+ model <- rpart(
+   Loan ~ Age + Income + Employment + Credit,
+   data = data,
+   method = "class"
+ )
+ print(model)
+ # 15
+ node(), split, n, loss, yval, (yprob)
+ * denotes terminal node
+
+ 1) root 15 6 yes (0.4000000 0.6000000) *
+
+
+ rpart.plot(model,
+   main = "Decision Tree for Loan Approval")
+

```



Result

A Decision Tree classifier was successfully constructed using **Age, Income, Employment Status and Credit Score** to predict **Loan Approval (Yes/No)**.

The tree can then be used to classify new customers based on their attributes.

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

ANS:

Compare Class A and Class B

The assessment provides the following two Year 9 class results.

Data

Class A

76, 35, 47, 64, 95, 66, 89, 36, 84

Class B

51, 56, 84, 60, 59, 70, 63, 66, 50

R Code:

```
# Q4 - Compare Class A and Class B
classA <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
classB <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)
# Mean
mean(classA)
mean(classB)
# Median
median(classA)
median(classB)
# Range
range(classA)
range(classB)
# Difference in range
diff(range(classA))
diff(range(classB))
# Boxplot
boxplot(classA, classB,
        names = c("Class A", "Class B"),
        main = "Comparison of Class A and Class B",
        ylab = "Exam Marks")
```

Results

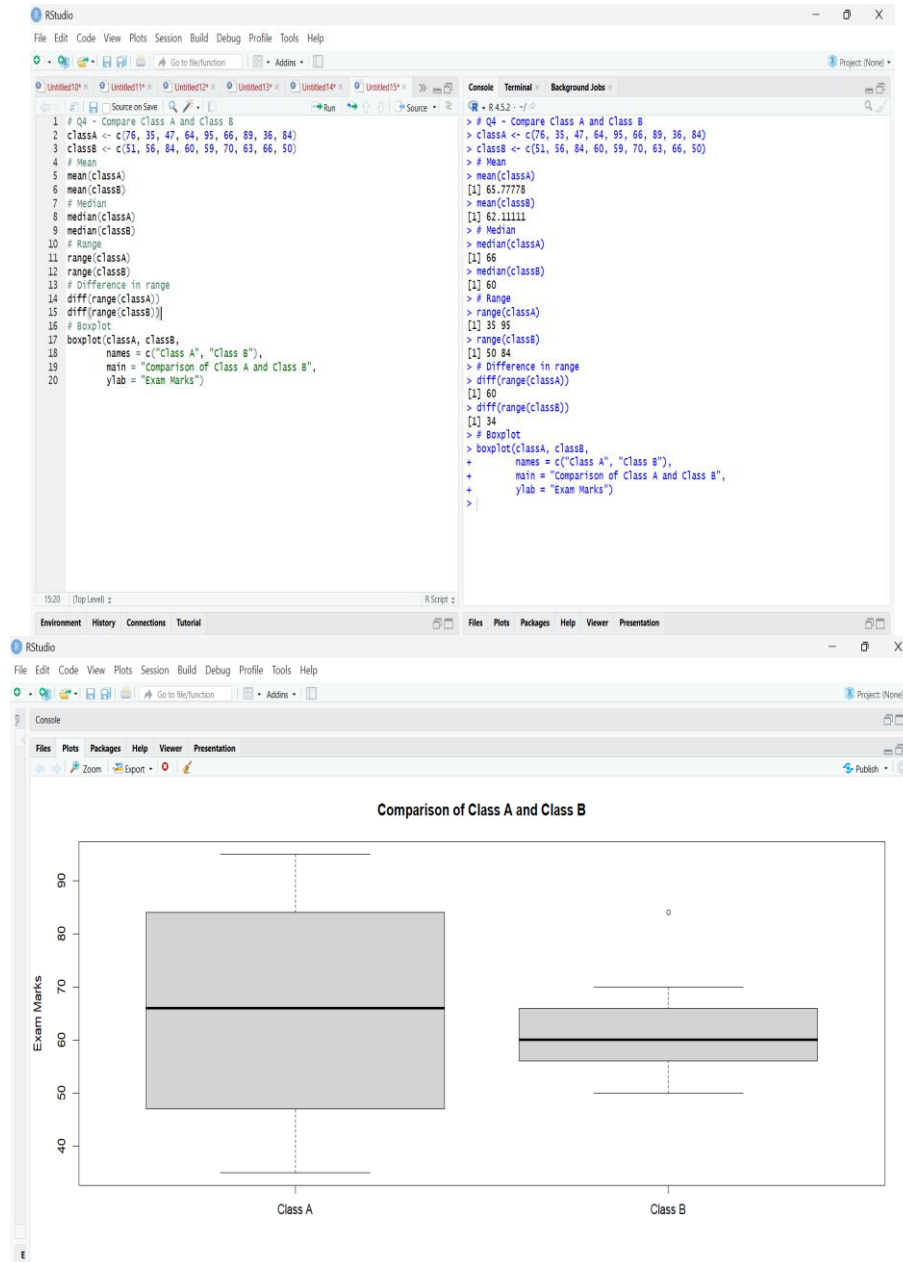
Statistic	Class A	Class B
Mean	65.78	62.11
Median	66	60
Range	60	34

Result

Class A has:

- Higher mean → **65.78**
- Higher median → **66**
- Larger range → **60**

Output:



Inference

Class A performed better on average because it has a higher mean and median. However, Class A has greater variation in marks, as indicated by its larger range. Class B has more consistent performance.

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

- Calculate the mean, median, and standard deviation of *age* and *%fat*.
- Draw the boxplots for *age* and *%fat*.
- Draw a *scatter plot* and a *q-q plot* based on these two variables.

Output:

The screenshot displays the RStudio interface with a script editor on the left and a console on the right. The script contains R code for data analysis, and the console shows the corresponding output.

```

1 #
2 # QS - AGE AND BODY FAT ANALYSIS
3 #
4 # Enter the data
5 age <- c(23, 23, 27, 27, 39, 41, 47, 49, 50,
6         52, 54, 54, 56, 57, 58, 58, 60, 61)
7 fat <- c(9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2,
8         34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7)
9 # Create data frame
10 data <- data.frame(Age = age, BodyFat = fat)
11 # Display data
12 data
13 #
14 # (a) MEAN, MEDIAN AND STANDARD DEVIATION
15 #
16 # Mean
17 mean(age)
18 mean(fat)
19 # Median
20 median(age)
21 median(fat)
22 # Standard deviation
23 sd(age)
24 sd(fat)
25 #
26 # (b) BOXPLOTS
27 #
28 # Boxplot for Age
29 boxplot(age,
30         main = "Boxplot of Age",
31         ylab = "Age")
32 # Boxplot for Body Fat
33 boxplot(fat,
34         main = "Boxplot of Body Fat",
35         ylab = "Body Fat (%)")
36 #
37 # (c) SCATTER PLOT
38 #
39 plot(age, fat,
40      main = "Scatter Plot of Age vs Body Fat",
41      xlab = "Age",
42      ylab = "Body Fat (%)",
43      pch = 19)
44 #
45 # CORRELATION
46 #
47 cor(age, fat)
48 #
49 # Q-Q PLOT
50 #
51 qqplot(age, fat,
52        main = "Q-Q Plot of Age and Body Fat",
53        xlab = "Age Quantiles",
54        ylab = "Body Fat Quantiles")
55 abline(0, 1)

```

The console output shows the following results:

```

Age BodyFat
1 23 9.5
2 23 26.5
3 27 7.8
4 27 17.8
5 39 31.4
6 41 25.9
7 47 27.4
8 49 27.2
9 50 31.2
10 52 34.6
11 54 42.5
12 54 28.8
13 56 33.4
14 57 30.2
15 58 34.1
16 58 32.9
17 60 41.2
18 61 35.7
> #
> # (a) MEAN, MEDIAN AND STANDARD DEVIATION
> #
> # Mean
> mean(age)
[1] 46.44444
> mean(fat)
[1] 28.78333
> # Median
> median(age)
[1] 51
> median(fat)
[1] 30.7
> # Standard deviation
> sd(age)
[1] 13.21862
> sd(fat)
[1] 9.254395
> #
> # (b) BOXPLOTS
> #
> # Boxplot for Age
> boxplot(age,
+         main = "Boxplot of Age",
+         ylab = "Age")
> # Boxplot for Body Fat
> boxplot(fat,
+         main = "Boxplot of Body Fat",
+         ylab = "Body Fat (%)")
> #
> # (c) SCATTER PLOT
> #
> plot(age, fat,
+      main = "Scatter Plot of Age vs Body Fat",
+      xlab = "Age",
+      ylab = "Body Fat (%)",
+      pch = 19)
> #
> # CORRELATION
> #
> cor(age, fat)
[1] 0.8176188
> #
> # Q-Q PLOT
> #
> qqplot(age, fat,
+        main = "Q-Q Plot of Age and Body Fat",
+        xlab = "Age Quantiles",
+        ylab = "Body Fat Quantiles")
> abline(0, 1)

```

